

THE DYNAMIC SLICING OPERATOR ON THE AM–GM CONE: A QUADRIC GEOMETRIC FRAMEWORK FOR KEPLERIAN ORBITS AND PERTURBED TRAJECTORIES

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ABSTRACT. We present a geometric and operator-theoretic formulation of central-force celestial mechanics using arithmetic mean–geometric mean (AM–GM) coordinates on factor surfaces. By mapping the turning points of an orbit (apoapsis r_a and periapsis r_p) to mean-value coordinates $T = (r_a + r_p)/2$, $X = (r_a - r_p)/2$, and $Y = \sqrt{r_a r_p}$, two-body Keplerian trajectories are represented as planar intersections of the Minkowski-like quadric cone $\mathcal{C} : X^2 + Y^2 = T^2$. We show that total energy E , specific angular momentum h , and central mass M isolate cleanly along orthogonal geometric directions of the cone. We then show that the $\mathfrak{so}(2, 1)$ symmetry algebra of $\mathcal{C}$ already contains everything needed to define time-dependent orbital evolution rigorously: the *dynamic slicing operator* $\hat{S}(t)$ is not an independent construction but the flow of a time-dependent element of $\mathfrak{so}(2, 1)$, and its two distinguished one-parameter subgroups are shown to be exactly energy-conserving eccentricity pumping and angular-momentum-conserving energy pumping. We then derive, from first principles (the standard perturbation identities $\dot{\epsilon} = \mathbf{v} \cdot \mathbf{a}_{\text{pert}}$ and $\dot{h} = r a_{\text{pert}, \theta}$), the explicit generator coefficients realizing atmospheric drag as a curve in this operator, and recover the classical circular-orbit decay rate as a special case.

1. INTRODUCTION

Classical two-body motion under a $1/r^2$ central force potential is conventionally parameterized via spatial state vectors $(\mathbf{r}, \mathbf{v})$ or osculating Keplerian orbital elements $(a, e, i, \omega, \Omega, \nu)$. While these representations excel at numerical integration, phase-space transitions between bound ($E < 0$), parabolic ($E = 0$), and hyperbolic ($E > 0$) regimes often require piecewise analytic parameterizations involving distinct trigonometric or hyperbolic anomalies.

In recent work on factor lattice geometry [1], an arithmetic mean–geometric mean coordinate transformation was introduced to organize binary factor pairs onto the quadric surface $X^2 + Y^2 = T^2$. In this paper, we extend this construction to continuous classical dynamics. By identifying factor pairs with radial turning points (r_a, r_p) , we demonstrate that all Keplerian trajectories manifest as plane slices of a single quadric cone, and that time evolution under external forces can be formalized as a continuous family of slicing operators $\hat{S}(t)$.

The version of this construction given here differs from an earlier draft in one important respect: rather than *postulating* $\hat{S}(t)$ as an independently-moving cutting plane $\alpha(t)X + \beta(t)T = \gamma(t)$, we *derive* it as the flow of a time-dependent element of the cone’s own isometry algebra $\mathfrak{so}(2, 1)$ (Section 5). This is not a cosmetic change: it guarantees, by construction and for every choice of time-dependent coefficients, that the evolving state remains exactly on $\mathcal{C}$ — i.e., that the instantaneous (X, Y, T) always correspond to an honest ellipse, at every instant, with no separate consistency check required. Section 6 develops this and then derives, from the standard perturbation equations of celestial mechanics, exactly which flow atmospheric drag realizes.

Key words and phrases. Kepler problem, AM–GM coordinates, Quadric cone, $\mathfrak{so}(2, 1)$ Lie algebra, Dynamic slicing operator, Orbital perturbations, Atmospheric drag.

2. THE ORBITAL AM–GM MAPPING

Consider a particle of mass m orbiting a central mass M in a Newtonian gravitational potential $V(r) = -\mu/r$, where $\mu = GM$. Let r_a and r_p denote the apoapsis and periapsis distances, respectively.

Definition 2.1 (Orbital Mean-Value Mapping). *The turning points (r_a, r_p) map to the coordinate tuple $(X, Y, T) \in \mathbb{R}^3$ via*

$$T = \frac{r_a + r_p}{2}, \quad X = \frac{r_a - r_p}{2}, \quad Y = \sqrt{r_a r_p}.$$

Direct substitution yields the fundamental algebraic identity

$$(1) \quad X^2 + Y^2 = \left(\frac{r_a - r_p}{2}\right)^2 + r_a r_p = \left(\frac{r_a + r_p}{2}\right)^2 = T^2.$$

Thus every physical realization of radial turning points lies on the cone surface $\mathcal{C} : X^2 + Y^2 = T^2$. The physical interpretation of these coordinates is immediate:

- $T = a$: semi-major axis (axial scale);
- $Y = b = a\sqrt{1 - e^2}$: semi-minor axis (transverse radius);
- $X = c = ae$: focal displacement (eccentricity displacement).

The cone constraint (1) is geometrically equivalent to the standard ellipse relation $(ae)^2 + b^2 = a^2$.

3. PHYSICAL INVARIANTS AS CONE COORDINATES

A major advantage of the AM–GM representation is that scalar physical invariants isolate along principal geometric axes.

Proposition 3.1 (Isolation of Physical Invariants). *For a two-body orbit on the cone $\mathcal{C}$:*

- (a) **Total specific energy** E is determined purely by the axial height T :

$$(2) \quad E = -\frac{\mu}{2T}.$$

Horizontal slices $T = \text{const.}$ represent constant-energy manifolds.

- (b) **Specific angular momentum** h is determined by the transverse coordinate Y and axial height T :

$$(3) \quad h = \sqrt{\frac{\mu Y^2}{T}} = \sqrt{\frac{\mu b^2}{a}}.$$

Hyperbolic cross-sections $Y = \text{const.}$ define manifolds of constant angular momentum.

- (c) **Central mass extraction.** Let v_a and v_p be the orbital speeds at apoapsis and periapsis. By angular momentum conservation ($h = v_a r_a = v_p r_p$), the central parameter $\mu = GM$ satisfies

$$(4) \quad \mu = GM = T \cdot v_a v_p \implies M = \frac{T \cdot v_a v_p}{G}.$$

Proof. (a) is the vis-viva energy formula $E = -\mu/(2a)$ rewritten with $T = a$. For (b): since $b^2 = a^2(1 - e^2) = r_a r_p$, the semi-latus rectum is $p = b^2/a = Y^2/T$, and the standard relation $h^2 = \mu p$ gives (3). For (c): $v_a = h/r_a$, $v_p = h/r_p$, so $v_a v_p = h^2/(r_a r_p) = h^2/Y^2$. Substituting $h^2 = \mu Y^2/T$ from (b) gives $v_a v_p = \mu/T$, i.e. $\mu = T v_a v_p$. $\square$

All three identities were checked numerically against a representative orbit ($\mu = 398600.4418 \text{ km}^3/\text{s}^2$, $r_a = 42164$, $r_p = 6800 \text{ km}$) and reproduce the standard vis-viva, angular-momentum, and mass-determination formulas to floating-point precision.

4. TRAJECTORY CLASSIFICATION

Theorem 4.1 (General Orbit Classification). *Let a family of orbital states satisfy the linear constraint $\alpha r_a + \beta r_p = \gamma$ for $\alpha, \beta, \gamma \in \mathbb{R}$. The resulting intersection with the cone $\mathcal{C}$ yields:*

- (1) **Elliptic/bound states** ($\alpha\beta > 0$): *the constraint slices both generators, forming a closed ellipse on $\mathcal{C}$ corresponding to $E < 0$ ($0 \leq e < 1$).*
- (2) **Parabolic/escape states** ($\alpha\beta = 0$): *the constraint is parallel to a boundary generator ray ($T = \pm X$), representing $E = 0$ ($e = 1$).*
- (3) **Hyperbolic/unbound states** ($\alpha\beta < 0$): *the constraint slices both nappes of $\mathcal{C}$, producing a hyperbola corresponding to $E > 0$ ($e > 1$).*

This is the direct specialization, to (r_a, r_p) in place of the generic factor pair (x, y) , of the General Conic Theorem of [1]. It also has a second, independent reading worth recording explicitly: a plane $\alpha X + \beta T = \gamma$ (no Y -dependence) cutting the cone of revolution $\mathcal{C}$ at various angles relative to the cone's opening half-angle (45° , since the cone's slope in the (X, T) -plane is 1) reproduces exactly the classical Apollonian definition of a conic section by a cutting plane: $|\alpha| < |\beta|$ gives an ellipse, $|\alpha| = |\beta|$ a parabola, $|\alpha| > |\beta|$ a hyperbola — consistent with the energy sign classification above via $E \propto -1/T$.

5. THE $\mathfrak{so}(2, 1)$ SYMMETRY ALGEBRA OF THE CONE

The isometry group preserving the cone $\mathcal{C}$, equivalently preserving the quadratic form $Q(X, Y, T) = X^2 + Y^2 - T^2$, is $SO(2, 1)$. We record its Lie algebra $\mathfrak{so}(2, 1)$ explicitly, in the ordered basis (X, Y, T) , since Section 6 will use every generator by name.

Definition 5.1 (Generators).

$$L = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad B_X = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad B_Y = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}.$$

Lemma 5.2 (Isometry and commutation relations). *Each of L, B_X, B_Y satisfies $G^\top \eta + \eta G = 0$ for $\eta = \text{diag}(1, 1, -1)$, i.e. each generates a one-parameter group of exact isometries of Q . They satisfy the standard $\mathfrak{so}(2, 1)$ commutation relations*

$$[L, B_X] = B_Y, \quad [L, B_Y] = -B_X, \quad [B_X, B_Y] = -L.$$

Proof. Direct matrix computation; verified symbolically and numerically. $\square$

Proposition 5.3 (Boost generator and rapidity). *The action of $\exp(sB_X)$ on an initial state (X_0, Y_0, T_0) yields*

$$(5) \quad \begin{pmatrix} X(s) \\ T(s) \end{pmatrix} = \begin{pmatrix} \cosh s & \sinh s \\ \sinh s & \cosh s \end{pmatrix} \begin{pmatrix} X_0 \\ T_0 \end{pmatrix}.$$

This boost preserves $Y = Y_0 = b$, leaving specific angular momentum h invariant, while scaling the semi-major axis $a(s) = Y_0 \cosh(s)$ and eccentricity $e(s) = \tanh(s)$. The rapidity parameter $s = \frac{1}{2} \ln(r_a/r_p)$ measures the logarithmic radial asymmetry of the orbit.

Both closed forms were checked numerically against a representative orbit: $s = \frac{1}{2} \ln(r_a/r_p)$ reproduces $\tanh(s) = e$ and $Y_0 \cosh(s) = T_0$ to floating-point precision.

Remark (Formal, not physical, continuation across $E = 0$). *Analytic continuation $s \rightarrow i\phi$ turns the hyperbolic boost (5) into the trigonometric relation $T(\phi) = Y_0 \cos(\phi)$, which is exactly the parametrization of the eccentric anomaly of a bound ellipse. This is a genuine algebraic correspondence between the two closed forms, worth recording as a structural curiosity of the algebra; it is not a physical process that continues one orbit into the other, since $E < 0$ and $E > 0$ orbits are*

not connected by any continuous physical evolution at fixed μ . We flag this explicitly to avoid the stronger reading.

Proposition 5.4 (Rotation generator: eccentricity pumping at fixed energy). *The action of $\exp(sL)$ on an initial state (X_0, Y_0, T_0) rotates (X, Y) while holding T exactly fixed:*

$$X(s) = X_0 \cos s - Y_0 \sin s, \quad Y(s) = X_0 \sin s + Y_0 \cos s, \quad T(s) = T_0.$$

Since $T = a$ is unchanged, $E = -\mu/(2T)$ is exactly conserved along this flow, while $X = ae$ and $Y = b$ — hence e and h — vary continuously.

This was verified numerically: propagating a representative orbit state under $\exp(sL)$ alone leaves T fixed to machine precision at every sampled s , while X, Y trace out the expected rotation.

Corollary 5.5 (Transitivity). *The identity component $SO(2, 1)_0$ acts transitively on $\mathcal{C} \setminus \{0\}$ (this is the classical fact that the future null cone in 2+1-dimensional Minkowski space is a single Lorentz orbit). Consequently, every physical orbit state is reachable from every other by some combination of L, B_X, B_Y ; the vertex $(0, 0, 0)$ is approached only as $s \rightarrow \infty$ along a boost, never reached in finite group parameter.*

6. THE DYNAMIC SLICING OPERATOR

6.1. Definition as a Lie-algebra flow. We now give the operative definition of the dynamic slicing operator, replacing the moving-plane heuristic with the flow it was standing in for.

Definition 6.1 (Dynamic Slicing Operator). *Given time-dependent coefficients $a(t), b(t), c(t)$, the dynamic slicing operator is*

$$(6) \quad \hat{S}(t) = \exp\left(\int_0^t [a(\tau)L + b(\tau)B_X + c(\tau)B_Y] d\tau\right),$$

and the instantaneous orbital state is $O(t) = \hat{S}(t) \cdot O(0)$, where $O(0) = (X_0, Y_0, T_0) \in \mathcal{C}$.

Because L, B_X, B_Y generate isometries of Q , $O(t) \in \mathcal{C}$ for every t automatically, for any choice of $a(t), b(t), c(t)$: the state is guaranteed to represent an honest ellipse at every instant, with no separate verification required. Definition 2 of the original construction — a plane $\alpha(t)X + \beta(t)T = \gamma(t)$ whose motion was linked only qualitatively to physical effects — is subsumed by (6): the two qualitative behaviors described there are exactly the two one-parameter subgroups of Section 5, made precise below.

6.2. The two distinguished flows.

Corollary 6.2. *Under $\hat{S}(t)$ generated by L alone: $\dot{\gamma}$ -type “operator tilting” is eccentricity pumping at exactly constant energy (Proposition 5.4). Under $\hat{S}(t)$ generated by B_X alone: “vertical translation” is angular-momentum-conserving energy exchange (Proposition 5.3); $\dot{s} > 0$ expands the orbit (low-thrust-type energy gain), $\dot{s} \rightarrow -\infty$ is impossible in finite time (energy loss under this pure flow asymptotically approaches, but never reaches, the circular orbit of radius Y_0).*

What remains is not a foundational question but a modeling one: given a real perturbing force, which $a(t), b(t), c(t)$ does it realize? We work this out from first principles for atmospheric drag.

6.3. Generator coefficients from a physical perturbation. Let $\mathbf{a}_{\text{pert}} = R\hat{r} + S\hat{\theta}$ be a perturbing acceleration in the orbital plane, decomposed into radial (R) and transverse (S) components. Two classical identities of perturbed two-body motion (see e.g. [4], [5]) give the instantaneous rate of change of specific energy $\varepsilon = v^2/2 - \mu/r$ and specific angular momentum h :

$$(7) \quad \dot{\varepsilon} = \mathbf{v} \cdot \mathbf{a}_{\text{pert}} = v_r R + v_\theta S, \quad \dot{h} = r S.$$

Since $T = -\mu/(2\varepsilon)$ and $Y = h\sqrt{T/\mu}$, the chain rule converts (7) directly into rates for the cone coordinates:

$$(8) \quad \dot{T} = \frac{2T^2}{\mu}(v_r R + v_\theta S),$$

$$(9) \quad \dot{Y} = \sqrt{\frac{T}{\mu}} r S + \frac{YT}{\mu}(v_r R + v_\theta S).$$

Both were checked numerically against the exact chain-rule computation from $\dot{\varepsilon}, \dot{h}$ directly, with agreement to floating-point precision, and (8)–(9) automatically satisfy the cone-tangency condition $X\dot{X} + Y\dot{Y} = T\dot{T}$ inherited from differentiating (1).

Since $\{L, B_X\}$ already span the two-dimensional tangent space to $\mathcal{C}$ at any point with $X \neq 0$ (i.e. any non-circular orbit), we may set $c(t) \equiv 0$ without loss of generality and solve (6)'s infinitesimal form

$$\dot{Y} = a(t)X, \quad \dot{T} = b(t)X$$

for the coefficients directly:

$$(10) \quad a(t) = \frac{\dot{Y}}{X}, \quad b(t) = \frac{\dot{T}}{X}.$$

6.4. Worked example: atmospheric drag. For drag opposing the velocity vector with ballistic coefficient $B = C_D A/m$ and local density $\rho(r)$,

$$\mathbf{a}_{\text{pert}} = -\frac{1}{2}\rho(r)Bv\mathbf{v} \implies R = -\frac{1}{2}\rho Bvv_r, \quad S = -\frac{1}{2}\rho Bvv_\theta.$$

Then $v_r R + v_\theta S = -\frac{1}{2}\rho Bv(v_r^2 + v_\theta^2) = -\frac{1}{2}\rho Bv^3$, and using $v_\theta r = h$:

$$rS = -\frac{1}{2}\rho Bv h.$$

Substituting into (8)–(9) and simplifying using $Y = h\sqrt{T/\mu}$ and the vis-viva identity $v^2 = \mu(2/r - 1/T)$ (which gives $1 + Tv^2/\mu = 2T/r$) collapses both rates to closed form:

$$(11) \quad \dot{T} = -\frac{\rho(r) B v^3 T^2}{\mu},$$

$$(12) \quad \dot{Y} = -\rho(r) B v Y \frac{T}{r}.$$

Both were checked numerically, at an arbitrary non-apsidal point on a representative orbit, against the exact formulas (8)–(9): agreement to floating-point precision.

Corollary 6.3 (Circular-orbit reduction). *For a circular orbit ($r = T = a$, $v^2 = \mu/T$), equation (11) reduces to*

$$\dot{T} = -\rho(T) B v T,$$

the classical atmospheric-decay rate for near-circular satellite orbits ([5]). This was verified numerically: the general formula (11), evaluated at a circular orbit, agrees with $-\rho Bva$ to floating-point precision.

From (10)–(12), the generator coefficients realizing atmospheric drag at any point of a non-circular orbit are

$$(13) \quad a(t) = -\frac{\rho(r) B v Y T}{r X}, \quad b(t) = -\frac{\rho(r) B v^3 T^2}{\mu X},$$

evaluated at the instantaneous (r, v) of the orbit. Equation (13) is exact and requires no further approximation beyond the standard drag force law; it may be evaluated pointwise along an orbit,

or averaged over one revolution (with $\rho(r(\nu))$ and $v(\nu)$ expressed via true anomaly ν) to obtain the usual secular decay rates, which we do not pursue further here.

Remark. *Both coefficients in (13) are manifestly negative for physical parameters ($\rho, B, v, Y, T, r, X > 0$ on a decaying elliptical orbit with $X > 0$), consistent with drag simultaneously reducing energy ($b(t) < 0$) and pumping eccentricity toward circularization at a rate coupled to it ($a(t) < 0$, reducing $X = ae$ relative to the shrinking $T = a$) — the well-known qualitative behavior of drag-induced circularization, now obtained as an exact consequence of (7) rather than asserted.*

7. CONCLUSION

The AM–GM conic representation provides a rigorous, computationally linear, and visually unified framework for celestial mechanics. Physical energy states, angular momentum, and central masses isolate into direct spatial dimensions on $X^2 + Y^2 = T^2$, and the classification of trajectories by energy sign is recovered as the classical Apollonian classification of plane sections of a cone. The dynamic slicing operator $\hat{S}(t)$, properly understood, is not a separate construction but the flow of a time-dependent element of the cone’s own $\mathfrak{so}(2, 1)$ symmetry algebra: its two distinguished one-parameter subgroups are exactly energy-conserving eccentricity pumping and angular-momentum-conserving energy exchange, and a real perturbing force is fully specified, with no further modeling freedom, once its instantaneous generator coefficients $a(t), b(t), c(t)$ are derived from the standard perturbation identities (7) — worked out here in closed form for atmospheric drag, and recovering the classical circular-decay rate as a consistency check.

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